

- J.L. Ba, J.R. Kiros, and G.E. Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.
- P. Baldi, P. Sadowski, and D. Whiteson. Searching for exotic particles in high-energy physics with deep learning. *Nature communications*, 5:4308, 2014.
- D. Barber and C.M. Bishop. Ensemble learning for multi-layer networks. In *Advances in neural information processing systems*, pp. 395–401, 1998.
- C. Blundell, J. Cornebise, K. Kavukcuoglu, and D. Wierstra. Weight uncertainty in neural network. In *International conference on machine learning*, pp. 1613–1622, 2015.
- T. Bui, D. Hernández-Lobato, J.M. Hernández-Lobato, Y. Li, and R.E. Turner. Deep Gaussian processes for regression using approximate expectation propagation. In *International conference on machine learning*, pp. 1472–1481, 2016.
- A. Damianou and N.D. Lawrence. Deep Gaussian processes. In *International conference on artificial intelligence and statistics*, pp. 207–215, 2013.
- A.G. De G. Matthews, M. Van Der Wilk, T. Nickson, K. Fujii, A. Boukouvalas, P. León-Villagrà, Z. Ghahramani, and J. Hensman. Gpflow: A Gaussian process library using tensorflow. *The Journal of Machine Learning Research*, 18(1):1299–1304, 2017.
- N. Durrande, D. Ginsbourger, and O. Roustant. Additive kernels for gaussian process modeling. *arXiv preprint arXiv:1103.4023*, 2011.
- D.K. Duvenaud, H. Nickisch, and C.E. Rasmussen. Additive Gaussian processes. In *Advances in neural information processing systems*, pp. 226–234, 2011.
- A. Esteva, B. Kuprel, R.A. Novoa, J. Ko, S.M. Swetter, H.M. Blau, and S. Thrun. Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542(7639):115–118, 2017.
- D. Flam-Shepherd, J. Requeima, and D. Duvenaud. Mapping Gaussian process priors to Bayesian neural networks. In *NIPS Bayesian deep learning workshop*, 2017.
- A.Y.K. Foong, Y. Li, J.M. Hernández-Lobato, and R.E. Turner. 'in-between' uncertainty in Bayesian neural networks. *ICML 2019 Workshop on Uncertainty and Robustness in Deep Learning*, 2019.
- A.Y.K. Foong, D.R. Burt, Y. Li, and R.E. Turner. On the expressiveness of approximate inference in Bayesian neural networks. In *Advances in neural information processing systems*, 2020.
- S. Fort, H. Hu, and B. Lakshminarayanan. Deep ensembles: A loss landscape perspective. *arXiv preprint arXiv:1912.02757*, 2019.
- Y. Gal. *Uncertainty in Deep Learning*. PhD thesis, University of Cambridge, 2016.
- Y. Gal and Z. Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *International conference on machine learning*, pp. 1050–1059, 2016.
- X. Glorot and Y. Bengio. Understanding the difficulty of training deep feedforward neural networks. In *International conference on artificial intelligence and statistics*, pp. 249–256, 2010.
- X. Glorot, A. Bordes, and Y. Bengio. Deep sparse rectifier neural networks. In *International conference on artificial intelligence and statistics*, pp. 315–323, 2011.
- T. Gneiting and A.E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- C. Guo, G. Pleiss, Y. Sun, and K.Q. Weinberger. On calibration of modern neural networks. In *International conference on machine learning*, pp. 1321–1330, 2017.
- D. Hafner, D. Tran, T. Lillicrap, A. Irpan, and J. Davidson. Noise contrastive priors for functional uncertainty. In *Uncertainty in Artificial Intelligence*, pp. 905–914, 2020.
- K. He, X. Zhang, S. Ren, and J. Sun. Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In *Proceedings of the IEEE international conference on computer vision*, pp. 1026–1034, 2015.

- J. Hensman, N. Fusi, and N.D. Lawrence. Gaussian processes for big data. In *Uncertainty in Artificial Intelligence*, pp. 282–290, 2013.
- J. Hensman, A.G. De G. Matthews, and Z. Ghahramani. Scalable variational Gaussian process classification. In *International conference on artificial intelligence and statistics*, pp. 351–360, 2015.
- D. Hernández-Lobato, J.M. Hernández-Lobato, and P. Dupont. Robust multi-class Gaussian process classification. In *Advances in neural information processing systems*, pp. 280–288, 2011.
- G.E. Hinton and D. Van Camp. Keeping the neural networks simple by minimizing the description length of the weights. In *Proceedings of the sixth annual conference on Computational learning theory*, pp. 5–13, 1993.
- G.E. Hinton, L. Deng, D. Yu, G.E. Dahl, A.R. Mohamed, N. Jaitly, A. Senior, V. Vanhoucke, P. Nguyen, T.R. Sainath, and B. Kingsbury. Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups. *IEEE Signal processing magazine*, 29(6): 82–97, 2012.
- S. Ioffe and C. Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *International conference on machine learning*, pp. 448–456, 2015.
- A. Kendall and Y. Gal. What uncertainties do we need in Bayesian deep learning for computer vision? In *Advances in neural information processing systems*, pp. 5574–5584, 2017.
- D.P. Kingma and J. Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- D.P. Kingma and M. Welling. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013.
- G. Klambauer, T. Unterthiner, A. Mayr, and S. Hochreiter. Self-normalizing neural networks. In *Advances in neural information processing systems*, pp. 971–980, 2017.
- A. Krizhevsky, I. Sutskever, and G.E. Hinton. Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems*, pp. 1097–1105, 2012.
- B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in neural information processing systems*, pp. 6402–6413, 2017.
- D.J.C. MacKay. A practical Bayesian framework for backpropagation networks. *Neural computation*, 4(3):448–472, 1992.
- W.J. Maddox, P. Izmailov, T. Garipov, D.P. Vetrov, and A.G. Wilson. A simple baseline for Bayesian uncertainty in deep learning. In *Advances in neural information processing systems*, pp. 13132–13143, 2019.
- T. Mikolov, K. Chen, G. Corrado, and J. Dean. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*, 2013.
- A. Mobiny, A. Singh, and H. Van Nguyen. Risk-aware machine learning classifier for skin lesion diagnosis. *Journal of clinical medicine*, 8(8):1241, 2019.
- V. Nair and G.E. Hinton. Rectified linear units improve restricted boltzmann machines. In *International conference on machine learning*, pp. 807–814, 2010.
- R.M. Neal. *Bayesian Learning for Neural Networks*. PhD thesis, University of Toronto, 1995.
- A. Nguyen, J. Yosinski, and J. Clune. Deep neural networks are easily fooled: High confidence predictions for unrecognizable images. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 427–436, 2015.

- K. Osawa, S. Swaroop, M.E.E. Khan, A. Jain, R. Eschenhagen, R.E. Turner, and R. Yokota. Practical deep learning with Bayesian principles. In *Advances in neural information processing systems*, pp. 4289–4301, 2019.
- T. Pearce, R. Tsuchida, M. Zaki, A. Brintrup, and A. Neely. Expressive priors in Bayesian neural networks: Kernel combinations and periodic functions. In *Uncertainty in Artificial Intelligence*, 2019.
- J. Postels, F. Ferroni, H. Coskun, N. Navab, and F. Tombari. Sampling-free epistemic uncertainty estimation using approximated variance propagation. In *Proceedings of the IEEE international conference on computer vision*, pp. 2931–2940, 2019.
- J. Ren, P.J. Liu, E. Fertig, J. Snoek, R. Poplin, M. Depristo, J. Dillon, and B. Lakshminarayanan. Likelihood ratios for out-of-distribution detection. In *Advances in neural information processing systems*, pp. 14680–14691, 2019.
- T. Salimans and D.P. Kingma. Weight normalization: A simple reparameterization to accelerate training of deep neural networks. In *Advances in neural information processing systems*, pp. 901–909, 2016.
- H. Salimbeni and M. Deisenroth. Doubly stochastic variational inference for deep Gaussian processes. In *Advances in neural information processing systems*, pp. 4588–4599, 2017.
- A. Shekhovtsov and B. Flach. Feed-forward propagation in probabilistic neural networks with categorical and max layers. In *International conference on learning representations*, 2018.
- J. Shi, S. Sun, and J. Zhu. A spectral approach to gradient estimation for implicit distributions. In *International conference on machine learning*, pp. 4644–4653, 2018.
- J. Snoek, Y. Ovadia, E. Fertig, B. Lakshminarayanan, S. Nowozin, D. Sculley, J. Dillon, J. Ren, and Z. Nado. Can you trust your model’s uncertainty? evaluating predictive uncertainty under dataset shift. In *Advances in neural information processing systems*, pp. 13969–13980, 2019.
- S. Sun, G. Zhang, J. Shi, and R. Grosse. Functional variational Bayesian neural networks. In *International conference on learning representations*, 2019.
- M. Titsias. Variational learning of inducing variables in sparse Gaussian processes. In *International conference on artificial intelligence and statistics*, pp. 567–574, 2009.
- S. Urban and P. van der Smagt. Gaussian process neurons. <https://openreview.net/forum?id=By-IifZRW>, 2018. Accessed: 2020-05-15.
- H. Wang, X. Shi, and D.Y. Yeung. Natural-parameter networks: A class of probabilistic neural networks. *Advances in neural information processing systems*, pp. 118–126, 2016.
- S. Wang and C. Manning. Fast dropout training. In *International conference on machine learning*, pp. 118–126, 2013.
- F. Wenzel, K. Roth, B.S. Veeling, J. Świątkowski, L. Tran, S. Mandt, J. Snoek, T. Salimans, R. Jenatton, and S. Nowozin. How good is the bayes posterior in deep neural networks really? *arXiv preprint arXiv:2002.02405*, 2020.
- C.K.I. Williams and C.E. Rasmussen. *Gaussian processes for machine learning*, volume 2. MIT press Cambridge, MA, 2006.
- A. Wu, S. Nowozin, E. Meeds, R.E. Turner, Hernández-Lobato J.M., and A.L. Gaunt. Deterministic variational inference for robust bayesian neural networks. In *International Conference on Learning Representations*, 2019.
- J. Yao, W. Pan, S. Ghosh, and F. Doshi-Velez. Quality of uncertainty quantification for bayesian neural network inference. *arXiv preprint arXiv:1906.09686*, 2019.